AMIRHOSSEIN KHAJEPOUR

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SUMMARY

Machine Learning and Data Science researcher with experience in clinical predictive modeling (pre-liver transplant mortality) and Persian NLP (hate-speech detection). Skilled in building end-to-end ML pipelines from data preprocessing and feature engineering to model selection, 10-fold cross-validation, imbalance handling (SMOTE), and ensemble learning. Also experienced in backend systems development (FastAPI, PostgreSQL, WebSocket, embedded systems integration), geospatial tooling, and blockchain analytics on Ethereum Classic, including on-chain dataset construction (BigQuery/SQL) and ML-based attack/fraud detection.

EDUCATION

M.Sc. in Information Technology Engineering — Amirkabir University of Technology (Tehran Polytechnic)
Tehran, Iran | Sep 2021 – Jul 2024

GPA: 17.55/20

Master's Thesis: Improving Detection of Selfish-Mining Attacks on the Ethereum Classic Blockchain Network Using Machine Learning

B.Sc. in Computer Engineering — Shiraz University of Technology *Shiraz, Iran | Sep 2016 – Sep 2020*
GPA: 16.92/20

CORE INTERESTS & FOCUS AREAS

Machine Learning & Data Science | Clinical Predictive Modeling (Healthcare AI) | Persian NLP (Hate Speech Detection) | Backend & Systems Engineering | Blockchain Analytics | Fraud & Attack Detection | Smart Contracts & Web3 (Learning Projects)

TECHNICAL SKILLS

Programming & Querying (Proficient): Python, SQL

ML & Data Science (Proficient): scikit-learn, pandas, NumPy; supervised learning (classification/regression), ensembles (Random Forest, Gradient Boosting), feature engineering, preprocessing, hyperparameter tuning, 10-fold CV, class-imbalance handling (SMOTE)

Backend & Systems (Proficient): FastAPI, RESTful API design, WebSocket real-time communication, PostgreSQL, embedded-system data integration

Deep Learning / NLP (Working knowledge): PyTorch, HuggingFace Transformers; BERT-based fine-tuning, LSTM baselines; lightweight GPT-2-style fine-tuning

Blockchain Data & Tooling (Working knowledge): Google BigQuery (SQL feature tables), XGBoost; dataset construction for on-chain analytics

Blockchain Development (Familiar): Solidity, Remix, Truffle, Ganache, ERC-20 patterns

Web3 Stack (Familiar): Web3.js, MetaMask integration, Hardhat

Web & GIS (Familiar): JavaScript, HTML/CSS, raster & MBTiles map tile formats

Distributed Systems & Cryptography (Familiar): Distributed system concepts; consensus (PoW, PoS, PBFT, IBFT, HotStuff); threshold signatures, BLS aggregation, ring signatures

WORK EXPERIENCE

Research Assistant – Scientometrics & Research Evaluation (NLP) — Islamic World Science Citation Center (ISC)
Shiraz, Iran | 2025–Present

- Conducting an NLP hate-speech detection project on Persian social media text using transformers (BERT) and deep learning baselines (e.g., LSTM).
- Performing data preparation, model training/evaluation, and experiment tracking to support research reporting and publication-ready results.

Research Assistant – Medical AI — Shiraz University of Medical Sciences *Shiraz, Iran | 2024–2025*

- Developed machine learning models to predict pre-liver transplant mortality, including model comparison and cross-validation.
- Built preprocessing pipelines (missing values, leakage removal, feature selection) and handled class imbalance to improve reliability of clinical prediction.

Research Assistant – Cybersecurity (Blockchain ML) — Amirkabir University of Technology *Tehran, Iran | 2021–2024*

- Developed ML-based approaches for blockchain attack detection, including dataset preparation and feature engineering from transaction/network signals.
- Evaluated multiple classifiers and optimized models for robust detection performance.

WiFi Network Support (Remote) — Tehran Telecommunication *Shiraz, Iran | 2020–2021*

- Provided remote support for WiFi infrastructure, troubleshooting connectivity and stability issues and assisting users with configuration and incident resolution.

Programmer, Codidac Team — Shiraz University of Technology *Shiraz, Iran | 2018–2019*

- Developed and maintained web applications as part of the university programming team, collaborating on feature implementation, debugging, and deployment support.

PROJECTS

1. Real-Time Embedded Data Pipeline (FastAPI | PostgreSQL | WebSocket | Python)

- Built a FastAPI backend that interfaces with an embedded system to collect live sensor/location data.
- Streamed incoming data to connected clients in real time over WebSocket, while persisting every record to a PostgreSQL database for storage and downstream analysis.
- Handled end-to-end data flow from embedded device → API → real-time client updates → database, including data validation and type consistency between the client and the database schema.

2. Offline Map Tile Downloader (Python | Raster & MBTiles Export)

- Developed a Python tool to download and package map tiles for offline use in GIS and mapping applications.
- Implemented export to both raster tile formats and MBTiles, enabling offline access to map data without a live internet connection.

3. Persian Hate Speech Detection (NLP | Transformers + LSTM + GPT-2 Fine-Tuning) (In Progress)

- Developed an NLP pipeline for Persian hate-speech detection using Transformer-based BERT models, an LSTM baseline, and fine-tuning lightweight GPT-2-style models on a Persian dataset; compared architectures and tuned training settings for classification performance.

4. Pre-Liver Transplant Mortality Prediction (10-Fold CV | Classical ML & Ensembles) (In Progress)

- Predicted patient mortality before liver transplantation using 10-fold cross-validation, evaluating Random Forest, XGBoost, Gradient Boosting, MLP, AdaBoost, Logistic Regression, and Linear SVM, plus ensemble methods Voting-Top3 (RF+XGB+GB) and Voting-Top4 (RF+XGB+GB+MLP).

5. Ethereum Classic Fraud Detection (Supervised ML | Python)

- Conducted a fraud detection study on the Ethereum Classic transaction dataset using SVM, Random Forest, and KNN, including hyperparameter optimization and performance comparison across models.

6. Ethereum Classic On-Chain Dataset Construction (Google BigQuery | SQL)

- Collected and engineered Ethereum Classic blockchain data from Google BigQuery using SQL, then built a structured dataset (feature tables) for downstream machine learning and analytics tasks.

7. Clinical Data Preprocessing & Feature Selection (Shiraz Transplant Dataset | Python)

- Performed end-to-end preprocessing on a pre-liver transplant dataset: missing-value handling, removal of target leakage and unstable clinical variables, and feature selection, preparing a reliable ML-ready dataset.

8. Class Imbalance Handling for Mortality Classification (SMOTE | Scikit-learn/Imbalanced-learn)

- Addressed class imbalance in the mortality prediction task using SMOTE, improving minority-class learning and stabilizing model evaluation under cross-validation.

SELECTED BLOCKCHAIN LEARNING PROJECTS

- ERC-20 Token + Simple ICO (Solidity): Built an ERC-20 token and a basic ICO-style token sale flow in a local environment.

- DApp + MetaMask Integration (Web3.js | Ganache): Built a simple DApp on a local Ganache blockchain and connected it to MetaMask for transactions.
- Voting + TimeLock Smart Contracts (Solidity): Implemented a basic on-chain voting system and a TimeLock contract for scheduled asset release.

AREAS I'M ACTIVELY DEVELOPING

- Financial & Risk Analytics: Building on my classification, ensemble modeling, and fraud/risk-detection background, I'm keen to grow into financial risk and underwriting analytics — applying predictive modeling and data-driven decisioning to credit risk, P&L forecasting, and portfolio performance.
- Financial Modeling in Excel/SQL: Actively studying core P&L and financial modeling concepts to complement my existing data science and analytics toolkit.

TEACHING EXPERIENCE

- Blockchain Course Assistant | Amirkabir University of Technology | 2023 | 30 hours
- Data Structures Course Assistant | Shiraz University of Technology | 2019 | 30 hours
- Discrete Structures Course Assistant | Shiraz University of Technology | 2018 | 30 hours

SELECTED ACHIEVEMENTS

- Ranked 14th in the National M.Sc. Entrance Exam (Information Technology)
- Member, Iran's National Elites Foundation (Bonyad-e Melli-e Nokhbegan)
- Elite Conscription Program (Sarbaz-e Nokhbeh), Ministry of Science, Research and Technology (MSRT), Iran

LANGUAGES

English — Upper-intermediate (estimated ~IELTS Overall 6.5)